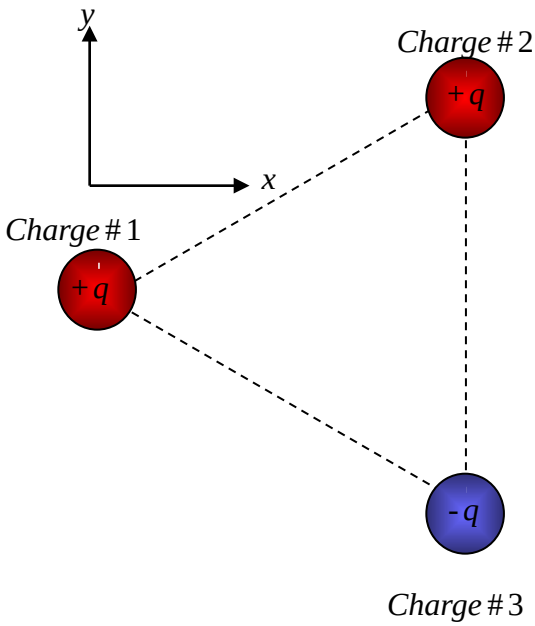
PH 1120B26BS01

Goals: Students will be able to calculate Coulomb's force between point charges.

1. Three charged particles lie at the vertices of an equilateral triangle as shown. All three particles have the same magnitude, but charges #1 and #2 are positive (+q) and charge #3 is negative (–q). What is the direction of the net electric force that charges #2 and #3 exert on charge #1?

(a) + x-direction
(b) - x-direction
(c) + y-direction
(d) - y-direction
(e) None of the above.



2. The magnitude of the electrostatic force between two charged particles is F. If the magnitude of each charge and their separation are doubled, what will be the new force between the charges?

(a) F. **(b)** $2F$ **(c)** $4F$ **(d)** $8F$ **(e)** $\frac{F}{2}$

3. Two charged particles are placed in the xy plane. $q_1 = +10.0\ nC$ at $(+3.00,\ 0.00)$ cm, and $q_2 = -5.00\ nC$ at $(-3.00,\ 0.00)$ cm.

(a) Calculate the force $\vec{F}_{12}$, exerted by the charge q_1 on the charge q_2. Sketch the force.

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(b) A third particle of charge $q_3 = -50.0\ nC$ is placed at $P(0.00,\ 4.00)$ cm. Calculate the net electrostatic force, $\vec{F}_{net}$ on q_3.

Practices to calculate the following unit vectors.

1. Directed from origin O(0,0)cm to point P(3,4)cm.

2. Directed from P(3,4)cm to origin O(0,0)

A challenging Question:

Two charged particles are placed in the xy plane. $q_1 = +10.0\ nC$ at $(+3.00,\ 0.00)$ cm, and $q_2 = 5.00\ nC$ at $(-3.00,\ 0.00)$ cm. A third charge $q_3 = 0.5\ nC$ is placed on the x-axis exactly in the middle between q_1 and q_2.

a) Which of the two charge have a greater force-influence over charge q_3.
b) Charge q3 is now moved into a position where the net electrostatic force acting on it is equal to zero. Should it be close to q_1 or q_2? Explain your answer.
c) Find the exact position where the net electrostatic force on charge q_3 is equal to zero.